

8. Demonstrate Express JS routing mechanism by developing two routes: User and Admin route. User route returns some user details and Admin route returns some admin details.

Program:

route.js

```
const express = require('express');

const app = express();

// User Route

app.get('/user', (req, res) => {

  res.json({ role: "User", name: "John Doe", status: "Active" });

});

// Admin Route

app.get('/admin', (req, res) => {

  res.json({ role: "Admin", name: "Super User", privileges: ["read", "write", "delete"] });

});

app.listen(3000, () => console.log('Routing server listening on port 3000'));
```

Execution steps:

1. Check the installation in terminal window by running, node -v and npm -v.
2. Create a project folder using, mkdir <dir_name> and use cd command to change the path to the current directory.
3. Initialize the project using, npm init -y.
4. Install Express using, npm install express.
5. Create a new file called route.js and type the above code.
6. Run the server using, node route.js
7. Now open browser and type in the url, <http://localhost:3000/user> and enter.
8. Follow the same for the admin route by typing in the url, <http://localhost:3000/admin>.

9. Observe the output for both the user and admin route being displayed on the browser.

9. Develop a React code to build a simple search filter functionality to display a filtered list based on the search query entered by the user.

Program:

search-app.js

```
import React, { useState } from 'react';

function App() {

  const [query, setQuery] = useState("");

  const items = ["Apple", "Banana", "Orange", "Mango", "Grapes"];

  const filteredItems = items.filter(item =>

    item.toLowerCase().includes(query.toLowerCase())

  );

  return (

    <div>

      <h2>Search Filter</h2>

      <input

        type="text"

        placeholder="Search fruits..."

        value={query}

        onChange={(e) => setQuery(e.target.value)}

      />

      <ul>

        {filteredItems.map((item, index) => <li key={index}>{item}</li>)}

      </ul>

    </div>

  );

}
```

```
</div>

);

}

export default App;
```

Execution steps:

1. Check the installation in terminal window by running, node -v and npm -v.
2. Open frontend folder and Create react app by using, npx create-react-app search-app in the terminal window.
3. Change the directory using cd command by typing, cd search-app.
4. From the explorer window, goto search-app folder/src/app.js and type the above code in it.
5. Now in the terminal window type, npm start.
6. Observe the output in browser window by giving the user text input in the search filter.